

Rice Yield Modeling Using Machine Learning Algorithms Based on Environmental and Agronomic Data of Pampanga River Basin, Philippines

Maria Cristina V. David

Department of Civil Engineering, Pampanga State Agricultural University, Philippines

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Abstract This study investigated the environmental and agronomic factors that influence rice crop yields in Pampanga River basin in Central Luzon Philippines. Specifically, this study examined the influence of fifteen (15) environmental and agronomic factors, to enumerate some: rainfall, temperature, wind speed, humidity, solar radiation, El Niño–Southern Oscillation (ENSO) classification, storms, tropical cyclones, rice crop yield (RCY), type of irrigation or water source; the data collected and investigated covered the years 2009 to 2018. Correlation and principal component analysis (PCA) were used to identify trends and patterns to identify significant environmental and agronomic variables that possibly affects rice crop yields. RCY was predicted using machine learning algorithms that include Linear Regression (LR), Artificial Neural Network (ANN) and Random Forest (RF); these models were evaluated using MSE (Mean Squared Error), MAE (Mean Absolute Error), and R^2 (Coefficient of Determination) metrics. Findings indicate that Correlation and PCA analysis identified average monthly rainfall (Rain-Ave) (0.387), total monthly rainfall (Rain-Total) (0.376), pressure at station (P-station) (0.388), and wind speed (Wind) (0.351) as among the environmental and agronomic factors that relatively have high loadings in the PCA analysis. Further, findings on the machine learning (ML) algorithms showed that RF model consistently performs across all evaluation methods, with lower MAE, MSE, and RMSE scores and higher R^2 scores

compared to the other models; the LR model also showed reasonably good performance in MAE and RMSE metrics, although it lagged behind the RF model; surprisingly the ANN model performed poorly despite it being the best performer during the model training phase. This study noted that overfitting is a major concern, and recommended that if a study has a limited data, other ML algorithms should be used instead of ANN (extensive data is required) as it could lead to overfitting, unsatisfactory results, poor generalization and performance on unseen data among other issues.

Keywords Crop Modeling, Environmental and Agronomic Data, Machine Learning Algorithms, Philippines

1. Introduction

Rice is a staple food for most Filipinos. Philippines is ranked 8th among the top rice producers in the world [1]. Yet, it is also the world's 2nd top rice importer [2]. High prices of agricultural inputs, typhoons, rising population, and diminishing land area planted to rice, have all played roles in setting the Philippines back in its rice self-sufficiency efforts.

The mean rice yield in the Philippines is higher than that

of Thailand, the world's biggest exporter of rice, where yields, over the last few years have been around 3 tons per hectare [3]. However, rice production in the Philippines is facing several challenges, including the increasing demand for rice, growing population, declining land area, environmental degradation, climate change, high cost of inputs, poor drainage and inadequate irrigation facilities. They are some of the major constraints to rice production. Climate change and the vulnerability of crop production to drought and heavy rainfall, especially during the typhoon season, severely affect rice production. The Philippines is affected by an average of twenty (20) typhoons coming in from the Pacific Ocean each year. Continuous heavy rains cause severe drainage problems in paddy fields, causing a significant reduction in rice yield and quality. Additionally, rainfed lowland rice paddies in the Philippines suffers from uncertain timing of arrival of rains, droughts, and submergences often in the same fields over the course of a single season or in different fields within a farm over the same season. Additionally, the high cost of inputs, predominantly fertilizer that deters farmers from applying optimum fertilizer amounts to input-responsive high-yielding varieties worsens the rice production difficulties.

These challenges and limitations to rice production in the Philippines prompted this study to investigate the environmental and agronomic factors that probably affect rice crop yields so as to develop rice crop production models that can help optimize resource use and improve rice production sustainability. This study investigated and analyzed the impact of fifteen (15) long-term environmental data and agronomic factors such as rainfall, temperature, wind speed, humidity, solar radiation, ENSO classification, number of storms, number of typhoons, rice crop yield, type of irrigation or water source, among others. Research and investigation of these factors through modeling can probably help identify trends and patterns that impact rice production and help optimize crop yield and quality. Several studies have been undertaken examining environmental and agronomic factors such as temperature, precipitation, and soil quality [4-5] and these studies have shown that high temperatures, drought, and soil salinity can reduce rice yields significantly; further adequate rainfall, nutrient-rich soil, and appropriate temperatures can increase rice yields significantly [6-7]. Agronomic factors such as fertilizer application, planting density, and irrigation practices can also impact rice production [8-9]. Adequate fertilizer application can improve soil fertility and nutrient uptake, resulting in increased rice yield. Planting density and irrigation

practices can also affect rice production, with studies indicating that optimal planting density and appropriate irrigation practices can lead to higher yields and water use efficiency [10].

This study, for exploratory data analysis, used correlation and principal component analysis in identifying trends, patterns, and significant environmental and agronomic variables that affected the rice production. Further, this study used three (3) methodologies in crop yield modeling: Artificial Neural Networks (ANN), Linear Regression (LR), and Random Forest (RF) to determine the possible influence of the fifteen (15) environmental and agronomic factors investigated in this study. Using ANN, LR, and RF in crop yield modeling is not new, for example, Guo et al. [11] were reported using ANN and Partial Least Squares Regression to predict rice yield in East China based on climate and agronomic traits data; also, Amaratunga et al. [12] estimated the paddy yield prediction using artificial neural network based on climatic data in Sri Lanka.

The objective of this study is to develop relevant crop production models using machine learning algorithms; models are useful in identifying the environmental and agronomic factors that affect rice production; models could led to the development of efficient and informed strategies in increasing rice crop yields and quality; models can provide insights and knowledge to policy makers in promoting sustainable agriculture particularly on rice. The study's findings may be useful to farmers, policymakers, and other stakeholders involved in rice production as possibly it can help optimize resource use while enhancing food security and livelihoods.

2. Materials and Methods

2.1. Study Site

Data in this study were collected from the Pampanga River basin in the Central Luzon, Philippines. Pampanga River basin has a catchment area of 10,434 km² covering eleven (11) provinces as shown in Figure 1. A substantial part of the study area (about 95%) are within the boundary of four (4) provinces: Nueva Ecija, Tarlac, Pampanga and Bulacan, thus only data from those provinces were included in the study. The climate of Pampanga River basin has distinct wet and dry seasons with annual precipitation of about 2,348 mm/year and allows for intensive rice paddy agriculture that contributes to about 30% of the rice production in the Philippines [13].

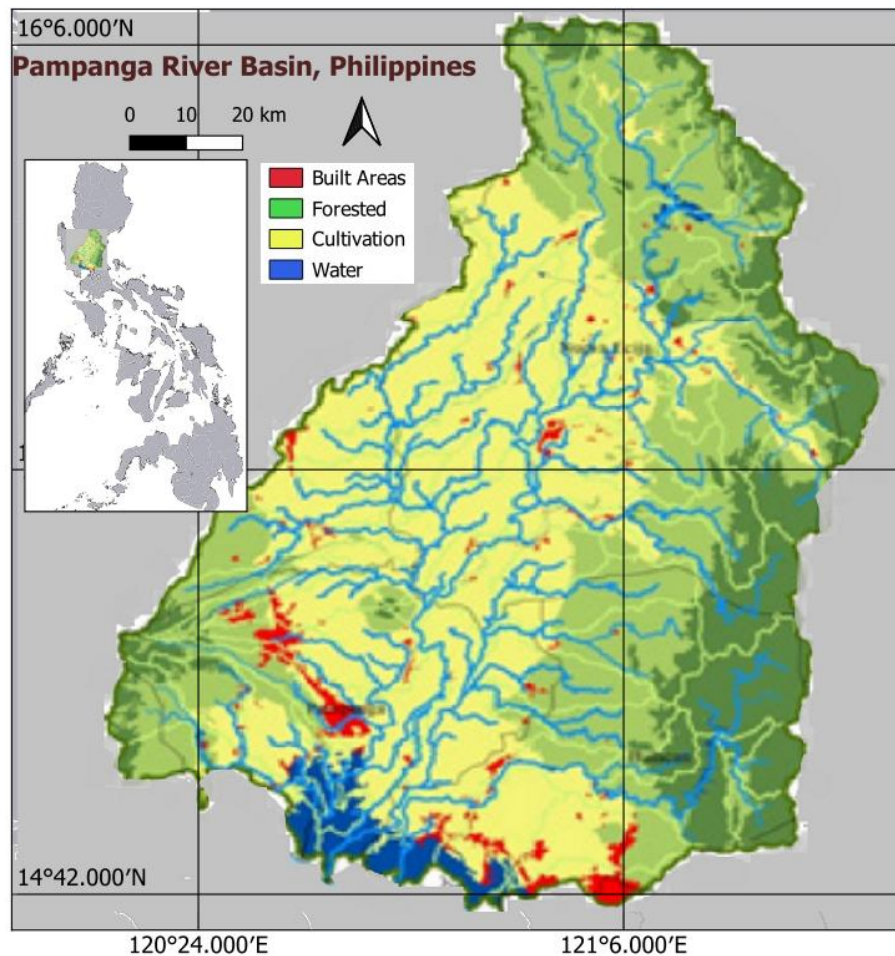


Figure 1. The study area: Pampanga River Basin in Central Luzon, Philippines (source: PAGASA)

2.2. Data Sources and Preparation

The historical data on climate factors, such as rainfall, temperature, relative humidity, solar radiation, wind speed, ENSO classification, and number of storms, typhoons, tropical depressions, were obtained from the National Meteorological and Hydrological Services' agency of the Philippines (PAGASA). The agronomic factors such as rice crop yields and water sources were obtained from Department of Agriculture (DA). The area irrigated and planted was obtained from National Irrigation Administration (NIA) – all from Region III offices in Central Luzon. The data sets were subjected to data cleaning, standardization, organization, coding, and pre-processing, including the imputation of missing values. Data Normalization was undertaken whenever necessary using the range values (0,1). The researcher employed the Orange Data Mining Package [14] in data analysis and modeling.

2.3. Correlation and Principal Component Analysis (PCA)

For exploratory data analysis, this study used

correlation and principal component analysis in identifying trends, patterns, and significant environmental and agronomic variables that affected the rice production. Correlation measures the statistical relationship, whether causal or not, between two random variables, correlation coefficient ranging between -1 and $+1$. However, principal component analysis (PCA) is a method used in large datasets containing a high number of dimensions/features per observation. It uses a technique of linearly transforming the data into a new coordinate system where the variation in the data can be described with fewer dimensions than the initial data. The principal components plot the data on two dimensions, enabling visual identification of clusters of closely related data points.

2.4. Artificial Neural Networks (ANN)

A multilayer perceptron was composed of an input layer of fifteen (15) features (Table 1) was developed. The first hidden layer had 20 neurons, and the second hidden layer had 10 neurons, and an output layer of one (1) neuron for Rice Crop Yield (RCY) (Figure 2).

A process of incrementally increasing the number of

neurons in both first and second hidden layers was subject to MSE (Mean Squared Error), MAE (Mean Absolute Error), and R^2 (Coefficient of Determination). The evaluation was used to identify the best combinations of neurons in both first and second hidden layers that result in a best fit to the data. After the best combinations of neurons for first and second hidden layers were determined, hyperparameter tuning was used to select the fit activation function and solver which was rectified linear activation function or ReLu and Stochastic Gradient Descent (SGD) for Solver. This study, as proposed by Kolmogorov [15], determined the number of neurons in the hidden layer by $N = 2m + 1$ where N is the neuron number in the hidden layer, and m is the neuron number

in the input layer.

2.5. Linear Regression (LR)

Linear regression models the relationship between the response (rice crop yield in MT per hectare) and explanatory variables (environmental and agronomic factors). It is described by the following equation:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_n X_n$$

where Y is the rice crop yield variable; α is the intercept; x_1 , x_2 , and $x_n, \dots$ represent the environmental and agronomic factors.

Table 1. Feature statistics of the Environmental and Agronomic Factors collected and analyzed in this study covering the years 2009 to 2018

Environmental and Agronomic Factors	Synonym	Unit	Mean	Min.	Max.	Dispersion
Rainfall-Average	Rain Ave	mm	6.52	0.00	27.28	0.98
Rainfall-Total	Rain Total	mm	176.27	0.00	738.70	1.05
Temperature-min	T-min	°C	23.38	19.22	26.23	0.06
Temperature-max	T-max	°C	31.54	28.41	35.25	0.05
Pressure-sea	P-sea	hPa	1105.84	1006.79	1965.64	0.26
Pressure-station	P-land	hPa	993.42	989.39	998.07	0.00
Relative Humidity	RH	%	74.53	57.63	88.26	0.10
Wind Speed	WS	m/s	2.45	1.82	3.74	0.14
Cloud Cover	CC	Okta	6.49	5.40	7.55	0.08
Sun	Sun	hrs/day	6.15	2.78	9.29	0.26
Southern Oscillation Index	SOI	index	0.29	-2.20	2.90	3.55
No. of Tropical Depressions	TD	n/yr	39.70	28.00	51.00	0.17
Number of Storms	Storms	n/yr	24.51	14.00	31.00	0.19
Number of Typhoons	Typhoons	n/yr	12.12	7.00	18.00	0.24
Super typhoons	ST	n/yr	5.10	1.00	9.00	0.47
Rice Crop Yield (RCY)	RCY	MT/ha	4.17	2.31	6.26	0.16

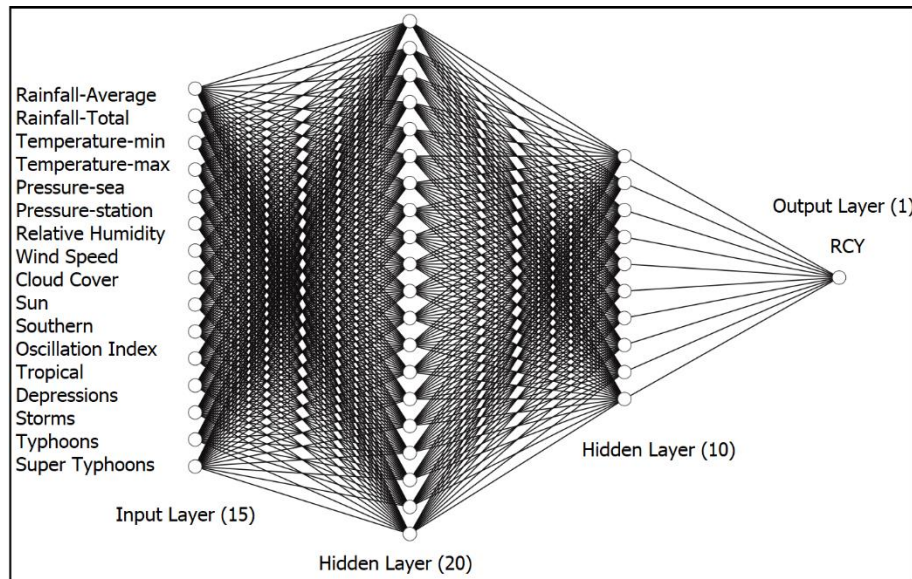


Figure 2. The Multilayer Perceptron architecture showing an input layer of 15 features, 20 neurons in the first hidden layer, 10 in the second hidden layer, and an output layer of one (1) neuron for RCY

2.6. Random Forest (RF)

Random Forest is a machine learning algorithm used for classification and regression problems that predict using an ensemble of decision trees, a technique that probably results in an accurate and substantial prediction model [16].

2.7. Model Evaluation

The evaluation of the models used the following metrics: MSE (Mean Squared Error), MAE (Mean Absolute Error), and R^2 (Coefficient of Determination) metrics. These metrics are calculated as follows:

$$MSE = (1/n) * \sum_i (x_i - y_i)^2$$

$$MAE = (1/n) * \sum_i |x_i - y_i|$$

where i runs from 1 to n , and $x_1 \dots x_n$ be the observed values and $y_1, \dots, y_n$ be the predicted values,

$$R^2 = 1 - RSS/TSS$$

where R^2 is a number between 0 and 1 and can be interpreted as the proportion of variation in the dependent variable that is predicted by the statistical model, RSS = sum of squares of residuals, and TSS = total sum of squares.

3. Results and Discussion

3.1. Correlation and Principal Component Analysis (PCA)

Figure 3 shows the Pearson correlation coefficient (r value) between two factors, Factor-1 and Factor 2, along

with their corresponding values. The Pearson correlation coefficient is a statistical measure that determines the strength and direction of the linear relationship between two variables. Weak correlations may imply that the relationship between the factors is not very strong, and it may be necessary to consider additional factors or perform further analysis to draw meaningful conclusions.

There are several weak positive correlations between Factor-1 (RCY) and various factors such as T-min, storms, T-max, Rain-Ave, Super typhoons, Rain-Total, Typhoons, RH, Cloud, and Sun. This suggests that there may be some degree of association between these factors and RCY. Conversely, there are weak negative correlations between Factor-1 (RCY) and factors like Wind, P-station, P-sea, SOI, and Tropical Depressions. These correlations indicate that as these factors increase, RCY tends to decrease to some extent. In terms of using these correlations for Principal Component Analysis (PCA), it can provide insights into the relationships between variables and help identify potential components with high intercorrelations.

Figure 4 and 5, also, Table 2 show the variance explained by each principal component (PC) in PCA, along with the loadings (coefficients) for the original variables on each PC. The variables included are Rain-Ave, Rain-Total, T-min, T-max, P-sea, P-station, RH, Wind, Cloud, and Sun. In PC1: this component explains 27.1% of the total variance. It has relatively high loadings for Rain-Ave (0.387), Rain-Total (0.376), P-station (0.388), and Wind (0.351). These variables contribute the most to PC1, and their signs (positive or negative) indicate the direction of their association with PC1. The PC2 component explains 19.6% of the total variance. It has loadings that are relatively close to zero for most variables. The variables with slightly higher loadings are Cloud (0.126) and Sun (-0.104), indicating a weak association with PC2.

Furthermore, PC3 explains 10.4% of the total variance. It has higher loadings for T-min (0.420) and T-max (0.550), suggesting a strong association with temperature variables. P-sea (-0.241) and RH (-0.305) also have relatively high

loadings, indicating an association with PC3. Further, PC4 explains 8.8% of the total variance; while PC5 explains 7.5% of the total variance. It has loadings that are relatively close to zero for most variables.

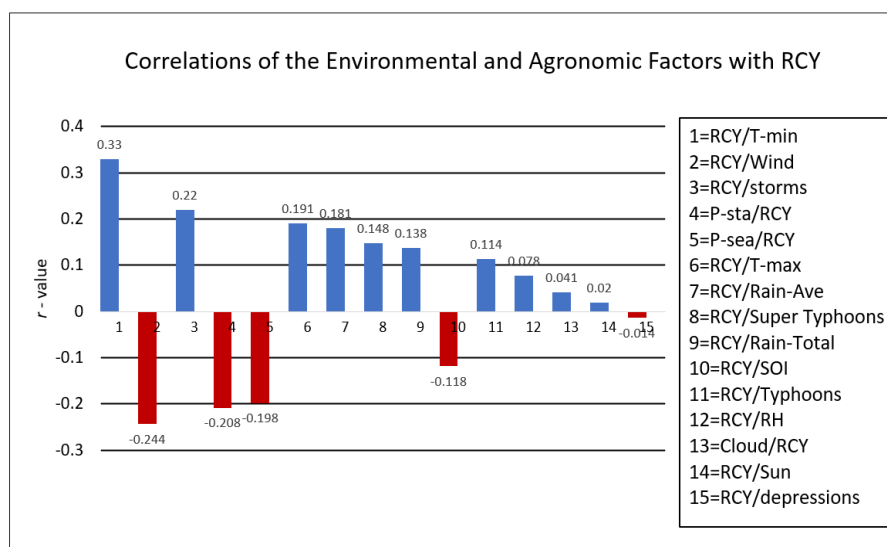


Figure 3. Correlations of the environmental and agronomic factors with RCY; T-min, Storms, T-max, and Rain-Ave are the relatively weakly positively correlated with RCY, while Wind, P-station, and P-sea are weakly negatively correlated with RCY

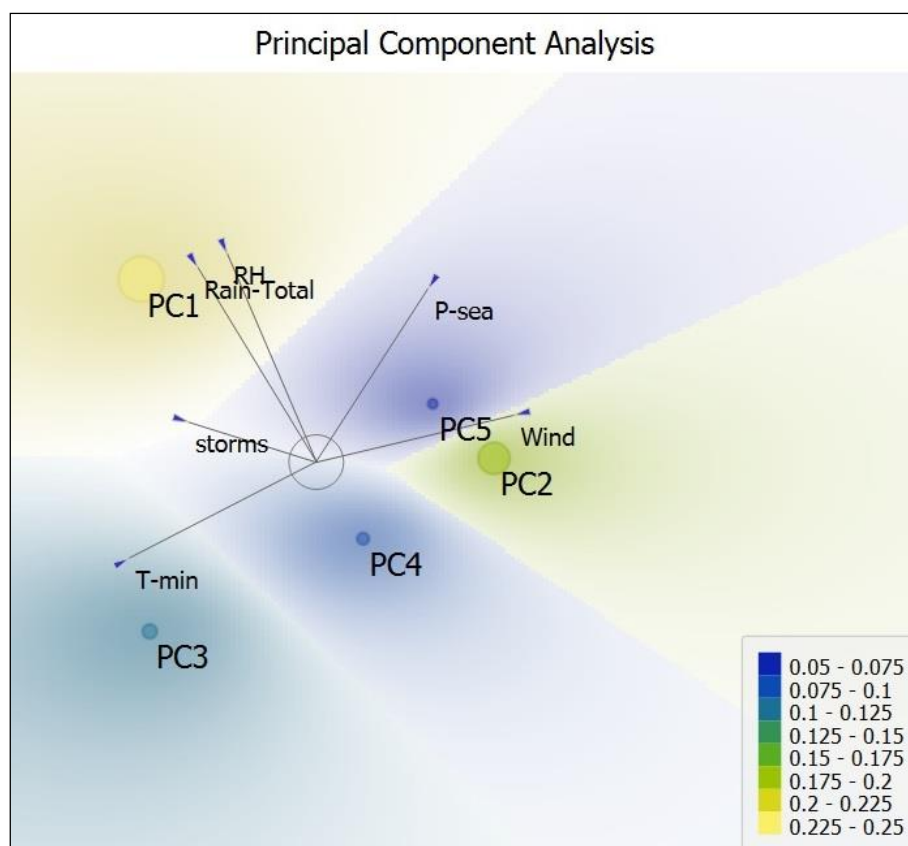


Figure 4. The five (5) principal components (PCA) illustrate the probable relationships of the environmental and agronomic factors in Pampanga River Basin

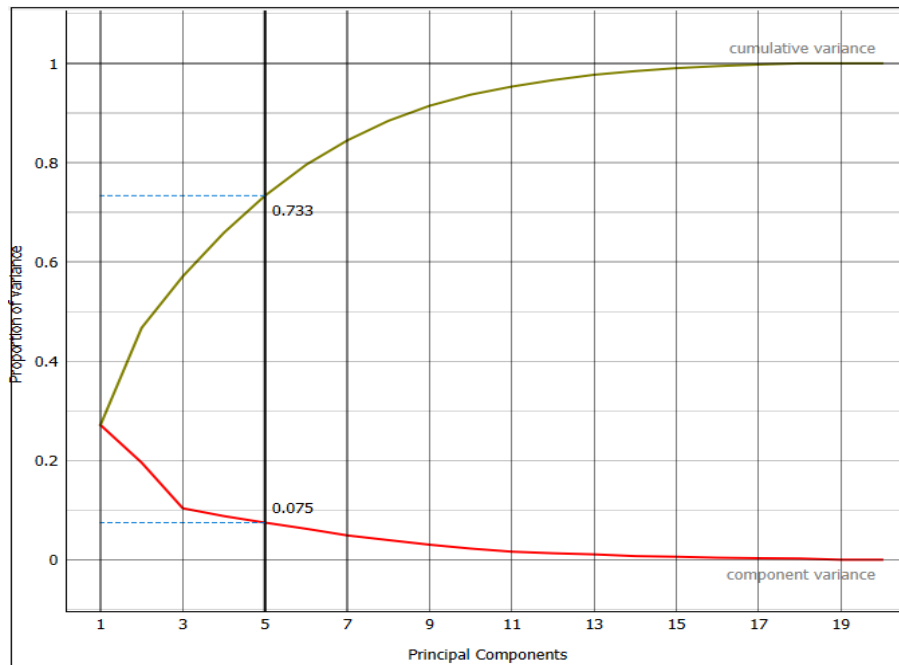


Figure 5. The five (5) principal components explain 73% of the variance of the data on environmental and agronomic factors relative to RCY in Pampanga River Basin

Table 2. PCA calculated variances of the environmental and agronomic factors relative to RCY

PCA	Variance	Rain-Ave	Rain-Total	T-min	T-max	P-sea	P-station	RH	Wind	Cloud	Sun
PC1	0.271	0.387	0.376	0.171	-0.156	0.016	-0.375	0.388	-0.214	0.351	-0.354
PC2	0.196	-0.013	-0.011	-0.226	-0.067	0.100	-0.020	-0.067	0.089	0.126	-0.104
PC3	0.104	0.017	-0.001	0.420	0.550	-0.241	-0.155	-0.132	-0.305	-0.108	0.263
PC4	0.088	0.068	0.039	0.075	0.114	-0.211	-0.009	0.020	-0.004	0.037	0.087
PC5	0.075	0.094	0.035	-0.114	-0.058	0.187	0.045	-0.009	0.103	0.059	-0.039

Con't. of **Table 2.** PCA calculated variances of the environmental and agronomic factors relative to RCY

PCA	Water Source =IP	Water Source =P	Water Source =RP	SOI	ENSO Class =EN	ENSO Class =LN	Tropical Depressions	Storms	Typhoons	Super Typhoons
PC1	-0.033	0.189	-0.157	0.021	-0.034	0.034	0.054	0.05	0.015	0.008
PC2	-0.002	-0.009	0.011	0.29	-0.402	0.402	-0.217	-0.32	-0.392	-0.43
PC3	0.165	0.069	-0.235	0.189	-0.154	0.154	0.14	0.138	-0.101	-0.167
PC4	0.089	0.13	-0.219	-0.054	0.364	-0.364	-0.521	-0.497	-0.249	0.01
PC5	0.777	-0.391	-0.387	-0.048	-0.02	0.02	0.028	0.012	0.087	0.053

Note: IP = irrigated palay, P=palay, RP=rainfed palay, EN= El Nino, LN= La Nina

In PCA, the variance explained by each component represents the amount of information captured by that component. For example, PC1 with relatively higher variances than the other PCs explain a larger proportion of the total variance in the environmental and agronomic

dataset. The loadings represent the contribution of each original variable to a particular component. Variables with higher loadings have a stronger influence on that component, and their signs indicate the direction of their association with the component.

3.2. Model Training, Testing, and Evaluation of ANN, LR, and RF Algorithms

3.2.1. Model Training

Figures 6, 7 and 8 illustrate the training performance of the three models used in this study: Linear Regression, Artificial Neural Network, and Random Forest. Table 3 provides the training performance scores for the three different models; performance scores were determined using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2) metrics. These performance scores provide insights into the accuracy and goodness-of-fit of the models on the training data.

Based on the training performance scores presented in Table 3, the findings are: (1) the Artificial Neural Network model shows the best overall performance among the three models, with the lowest MSE, RMSE, and MAE scores and the highest R^2 score. This suggests that the neural network model has achieved a better fit to the training data compared to the other models; (2) the Linear Regression model performs relatively worse than the other models, with higher MSE, RMSE, and MAE scores, and a lower R^2 score. This indicates that the linear regression model may

not capture the underlying patterns and relationships in the data as effectively as the other models; while the Random Forest model shows good performance, with relatively low MSE, RMSE, and MAE scores, and a high R^2 score. While not as strong as the Artificial Neural Network model, it still demonstrates a reasonable fit to the training data. It should be noted that these performance scores are based on the training data, and the models' true performance can be better assessed by evaluating them on unseen test data or using cross-validation techniques.

Table 3. Training performance score of the models; 70%, n= 168 data were used for model training

Model	Training Performance Score			
	MSE	RMSE	MAE	R^2
ANN	0.024	0.155	0.104	0.943
Linear Regression	0.190	0.436	0.350	0.551
Random Forest	0.032	0.179	0.133	0.924

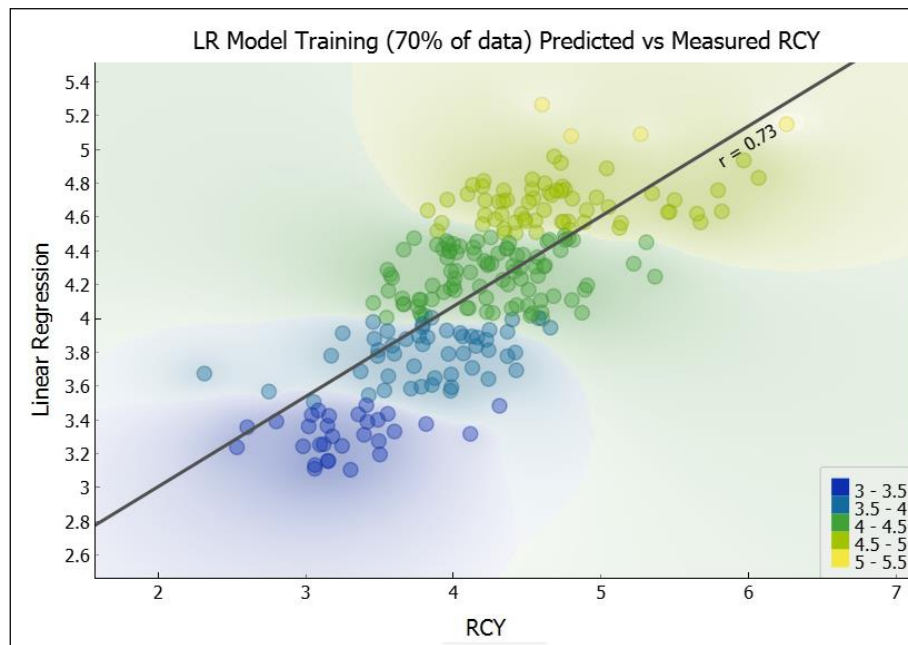


Figure 6. Training performance of the Linear Regression model showing the predicted and measured rice crop yields (metric ton per hectare), $r=0.73$

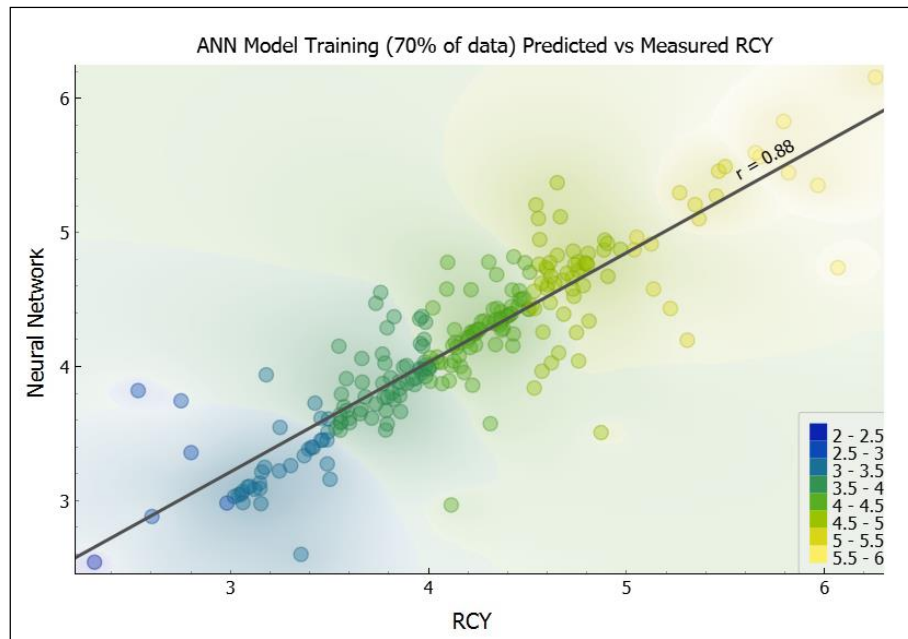


Figure 7. Training performance of the Artificial Neural Network model showing the predicted and measured rice crop yields (metric ton per hectare), $r=0.88$

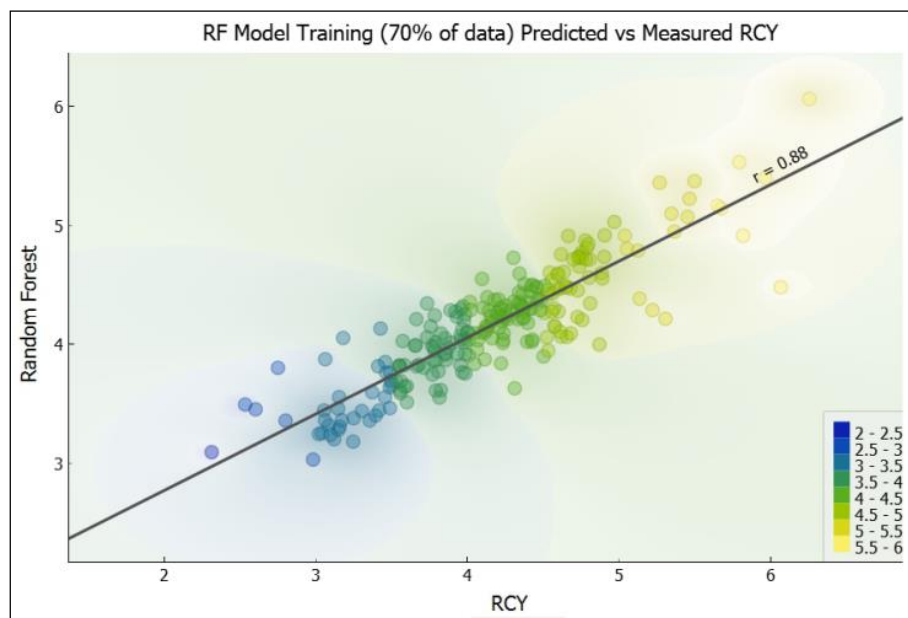


Figure 8. Training performance of the Random Forest model showing the predicted and measured rice crop yields (metric ton per hectare), $r=0.88$

3.2.2. Model Testing

Figures 9, 10, and 11 illustrate the testing performance of the three models; Table 4 provides the testing performance scores for the same three models: Artificial Neural Network, Linear Regression, and Random Forest. These testing performance scores provide insights into how well the models generalize and work to unseen data.

This study deduced, based on the testing performance presented in Table 4, the following: (1) Random Forest model performed the best among the three models in terms of testing performance. It has the lowest MSE, RMSE, and

MAE scores, indicating better accuracy and smaller deviations between predicted and actual values. The Random Forest model also has the highest R^2 score, suggesting it explains a larger proportion of the variance in the testing data compared to the other models; (2) Linear Regression model performed moderately with relatively low MSE, RMSE, and MAE scores, and a decent R^2 score. It demonstrates a reasonable fit to the testing data, although it is outperformed by the Random Forest model; (3) Surprisingly, despite being the best performer in the model training (Table 3), Artificial Neural Network showed the

poorest performance among the three models in terms of testing. It should be noted that the testing performance scores were based on evaluating the models on unseen data, providing an assessment of their generalization capabilities.

However, further analysis and comparison of the models on multiple evaluation metrics and validation techniques would provide a more comprehensive evaluation of their performance.

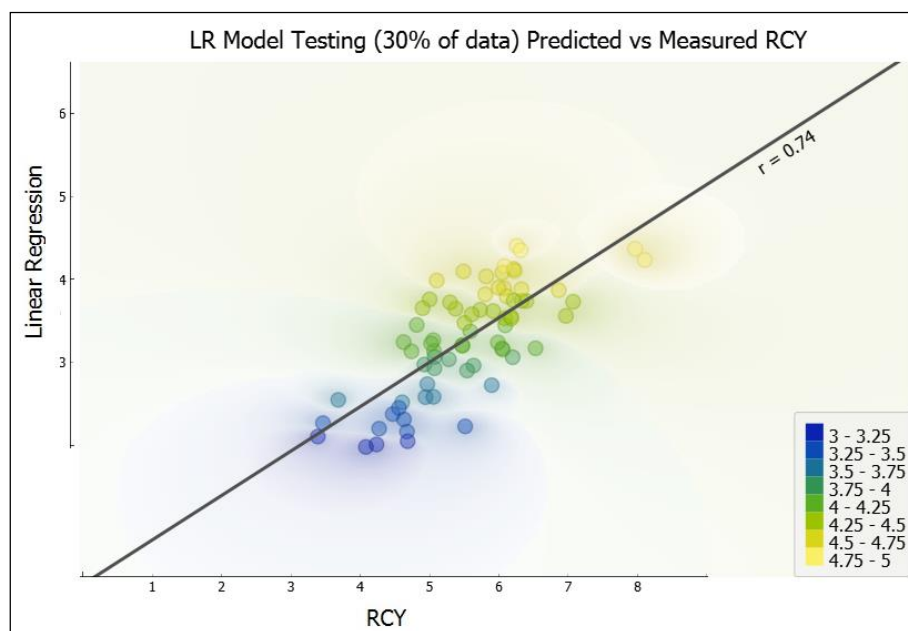


Figure 9. Testing performance of the Linear Regression model showing the predicted and measured rice crop yields (metric ton per hectare), $r=0.74$



Figure 10. Testing performance of the Artificial Neural Network model showing the predicted and measured rice crop yields (metric ton per hectare), $r=0.65$

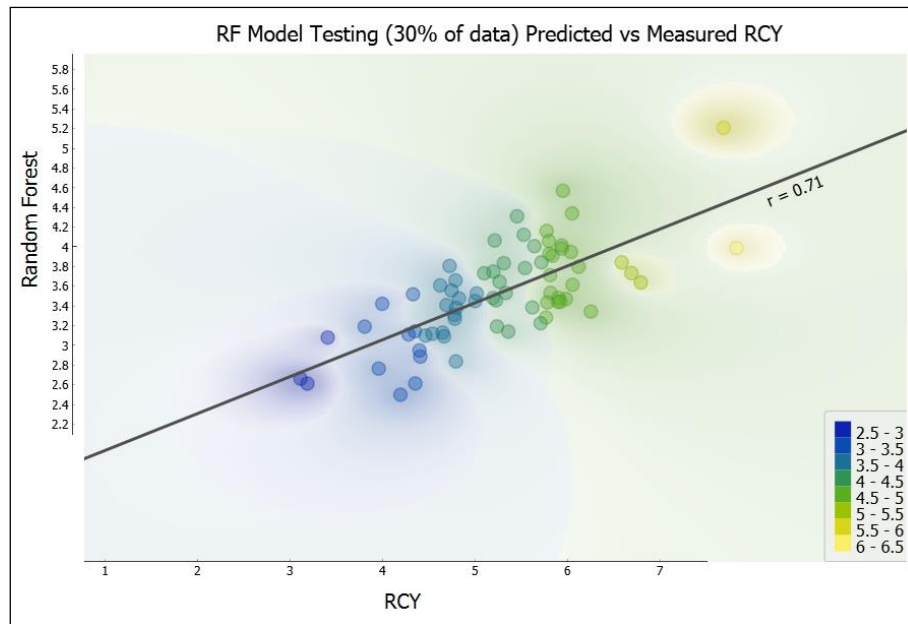


Figure 11. Testing performance of the Random Forest model showing the predicted and measured rice crop yields (metric ton per hectare), $r=0.71$

Table 4. Testing performance score of the models, thirty percent (30%, $n=72$) of the data were used for model testing

Model	Testing Performance Score			
	MSE	RMSE	MAE	R ²
ANN	0.287	0.536	0.426	0.365
Linear Regression	0.194	0.440	0.349	0.572
Random Forest	0.159	0.399	0.295	0.647

3.2.3. Model Evaluation

Table 5 provides the evaluation scores for the models (Artificial Neural Network - ANN, Linear Regression - LR, Random Forest - RF) using different evaluation methods: Leave One Out (LOO), Cross Validation (CV) with 5-fold, and Random Sampling (RS). The evaluation metrics used are Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R²).

The scores in Table 5 provide insights into the performance of the models using different evaluation methods. Based on these scores, the researcher reports the following observations: (1) Random Forest (RF) model consistently performs well across all evaluation methods, with lower MAE, MSE, and RMSE scores and higher R² scores compared to other models; (2) Linear Regression (LR) model also shows reasonably good performance, particularly in terms of MAE and RMSE, although it slightly lags behind the Random Forest model; and lastly, Artificial Neural Network (ANN) model generally performs relatively poorer compared to other models, as indicated by higher MAE, MSE, and RMSE scores and lower R² scores.

Table 5. Evaluation scores of the models (out of sample testing) using MAE, MSE, RMSE, and R² metrics; methods used were Leave one out (LOO), Cross Validation (CV, 5-fold), and Random Sampling (RS)

Metric	Method	ANN	LR	RF
MAE	LOO	0.539	0.383	0.338
	CV	0.396	0.378	0.339
	RS	0.437	0.385	0.336
MSE	LOO	0.474	0.227	0.181
	CV	0.251	0.224	0.191
	RS	0.306	0.227	0.188
RMSE	LOO	0.668	0.476	0.425
	CV	0.501	0.473	0.437
	RS	0.553	0.476	0.434
R ²	LOO	0.217	0.464	0.537
	CV	0.408	0.472	0.549
	RS	0.249	0.442	0.537

Why is the Artificial Neural Network (ANN) model, despite showing the best performance during the model training, performed poorly in the model testing and evaluation phase?

The researcher speculates that it could be probably attributed to several factors, possibly to the following: (1) Overfitting. This occurs when a model becomes too complex and starts to memorize the training data instead of learning the underlying patterns [17-18]. If the ANN model is overfitting the training data, it will perform well on the training set but poorly on the testing and evaluation sets. Overfitting can be caused by a model with too many layers or neurons, insufficient regularization techniques, or limited training data [17]; (2) Insufficient data, if the amount of training data is limited, which is the case in this study, the ANN model might not have enough examples to learn the underlying patterns effectively, resulting in poor generalization and performance on unseen data [19]; (3) Hyperparameter tuning as the performance of an ANN model can be highly sensitive to the choice of hyperparameters such as the learning rate, the number of hidden layers, activation functions, regularization techniques, among other factors [20]. If the hyperparameters are not properly tuned, it can lead to suboptimal performance on new data [19-20].

4. Conclusions

For machine learning model development, Random Forest (RF) model consistently performs well across all evaluation methods, with lower MAE, MSE, and RMSE scores and higher R^2 scores compared to the other models; also, Linear Regression (LR) model shows reasonably good performance.

From the PCA analysis and correlation analysis combined, the researcher has drawn the following conclusions: (1) The correlation analysis provided insights into the strength and direction of linear relationships between variables. It indicated weak positive correlations between RCY (Factor-1) and variables such as T-min, Storms, T-max, Rain-Ave, Super Typhoons, Rain-Total, Typhoons, RH, Cloud, and Sun. On the other hand, there were weak negative correlations between RCY and variables like Wind, P-station, P-sea, SOI, and Tropical Depressions. These correlations suggest potential associations between these factors and RCY, albeit with weak linear relationships; (2) PCA analysis revealed the variance explained by each principal component (PC) and the loadings of variables on each PC. It indicated that PC1 explained the largest amount of variance (27.1%) and was primarily associated with Rain-Ave, Rain-Total, P-station, and Wind. PC2 explained the second-largest variance (19.6%) and showed associations with Cloud and Sun. PC3 and PC4 explained 10.4% and 8.8% of the variance, respectively, with PC3 being strongly associated with temperature variables (T-min and T-max) and PC4 having relatively small loadings for most variables. PC5 explained 7.5% of the variance and showed associations with RH and Wind; (3) There are multiple weak positive correlations between RCY and factors such as temperature, rainfall,

typhoons, and atmospheric conditions (RH, Cloud, Sun). These associations are also reflected in the PCA results, where variables related to rainfall (Rain-Ave, Rain-Total) and weather conditions (Wind, Cloud, Sun) contribute significantly to the explained variance in PC1. Additionally, PC3, which is associated with temperature variables, aligns with the weak positive correlation between RCY and T-max. The negative correlations found between RCY and factors like Wind, P-station, P-sea, SOI, and Tropical Depressions are also reflected in the PCA results, with these variables having smaller or even negative loadings on various PCs.

Recommendations

The researcher recommends, with regard to the issue of Artificial Neural Network (ANN) model, where despite it showing the best performance during the model training, it performed poorly in the model testing and evaluation phase. To address this concern, maybe, the following should be considered: (1) address the issue of overfitting for this might be the reason, (2) if the study has limited data. Maybe, other machine learning algorithms should be used instead of Artificial Neural Network as lack of data for both training and testing would also lead to overfitting. Besides, insufficient data may lead the ANN to not having enough examples and data to learn the underlying patterns effectively, leading to unsatisfactory results, to poor generalization and performance on unseen data [17-18] among other issues; (3) future studies should investigate the potential benefits of combining multiple machine learning algorithms including ensemble methods, convolutional neural networks (CNNs) [21], SVM (support vector machines) [22], recurrent neural networks (RNNs) and related algorithms attentive on identification of methods to mitigate the risk of overfitting and improve predictive accuracy of ML algorithms. Certainly, more sophisticated neural network architectures beyond what's been mentioned in this study should also be explored, including transformers that may offer improved performance and accuracy.

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